QUANTUM SIMULATION

# Floquet Hamiltonian engineering of an isolated many-body spin system

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Controlling interactions is the key element for the quantum engineering of many-body systems. Using time-periodic driving, a naturally given many-body Hamiltonian of a closed quantum system can be transformed into an effective target Hamiltonian that exhibits vastly different dynamics. We demonstrate such Floquet engineering with a system of spins represented by Rydberg states in an ultracold atomic gas. By applying a sequence of spin manipulations, we change the symmetry properties of the effective Heisenberg XYZ Hamiltonian. As a consequence, the relaxation behavior of the total spin is drastically modified. The observed dynamics can be qualitatively captured by a semiclassical simulation. Engineering a wide range of Hamiltonians opens vast opportunities for implementing quantum simulation of nonequilibrium dynamics in a single experimental setting.

According to Floquet's theorem (*1*), the stroboscopic dynamics of periodically driven quantum systems is effectively described by a time-independent Hamiltonian that can be engineered by controlling the properties of the drive (*2*, *3*). This type of Floquet Hamiltonian engineering (FHE) leads to the observation of unconventional phases of matter (*4*, *5*), dynamical phase transitions (*6*), lattice gauge theories (*7*) and Floquet-topological matter (*8*, *9*). A paradigmatic example of FHE involves a series of spin echo sequences (*10*, *11*) applied to a system of spins embedded in an inhomogeneous environment, as encountered, for example, in nuclear magnetic resonance (NMR). The pulse sequence inverts the spins multiple times such that they finally return to their initial states and appear to be effectively decoupled from the environment. In a Hamiltonian description, this corresponds to the emergence of a vanishing Hamiltonian, that is, the effective time-independent Hamiltonian over the spin echo sequence becomes zero. Although this type of sequence is suitable to decouple the dynamics of single spins, more sophisticated sequences have been used to decouple two-body interactions (*12*) and to engineer Hamiltonians for studying many-body localization of mixed quantum states (*13*).

The application of multipulse protocols, as, for example, introduced by (*14*, *15*), to closed quantum systems facilitates the quantum engineering of many-body spin systems. Owing to the genuine decoupling from the environment and the high degree of control down to the single-particle level, the system's universality classes, symmetries, and type of interactions become tunable. In this study, we demonstrate the realization of such programmable many-body spin Hamiltonians and observe the modification of the out-of-equilibrium dynamics as a function of the target Hamiltonian's parameters. We use an ultracold atomic Rydberg gas which can be well decoupled from the environment and has already been established as an excellent platform for quantum simulation of closed systems (*16*, *17*) enabling, for example, the preparation of pure many-body quantum states in random (*18*) and controllable spatial geometries (*19*, *20*). The unitary dynamics of this system are captured by spin models (*21*), which serve as prototypical templates for studying quantum magnetism emerging from different classes of Hamiltonians (*22*).

The key idea of this work is to transform the naturally given spin Hamiltonian $H_{\mathrm{Nat}}$ into a target Floquet Hamiltonian $H_{\mathrm{Floq}}$ using a sequence of periodically applied control pulses (Fig. 1A). To demonstrate programming of the resulting effective Hamiltonian, we engineer a tunable Heisenberg XYZ Hamiltonian from a naturally given XX Hamiltonian. We benchmark FHE by comparing the emerging dynamics of the driven system with the one expected from an effective time-independent Hamiltonian. As a demonstration of tunability, we modify the symmetries in an effective XYZ Heisenberg model and analyze the resulting change of the out-of-equilibrium relaxation dynamics.

More specifically, we consider a Heisenberg XX Hamiltonian with an external driving field

$$H(t) = H_{\mathrm{XX}} + H_{\mathrm{drive}}(t) \qquad (1)$$

where $H_{\mathrm{XX}} = \sum_{i,j} J_{ij}/\hbar \left( S_x^i S_x^j + S_y^i S_y^j \right)$ and $H_{\mathrm{drive}}(t) = \sum_i \Omega(t) \left[ \cos\phi(t) S_x^i + \sin\phi(t) S_y^i \right]$, with $\hbar$ being the reduced Planck's constant. Here, $S_\alpha^i$ ($\alpha \in x, y, z$) are spin-1/2 operators, $\Omega(t)$ is the time-dependent Rabi frequency of the drive, $\phi(t)$ is the phase, and $J_{ij}$ is the interaction coefficient between spins $i$ and $j$. As schematically depicted in Fig. 1B, we use the drive $H_{\mathrm{drive}}(t)$, which consists of a periodic sequence of four global $\pi/2$ pulses realizing different spin operators $(S_x, -S_y, S_y, -S_x)$; the pulses are separated by delay times $\tau_1 = \tau(1-2v+2w), \tau_2 = \tau(1+2u-2w)$, and $2\tau_3 = 2\tau(1-2u+2v)$, where $u$, $v$, and $w$ are dimensionless parameters. Under the periodic driving $H_{\mathrm{drive}}(t)$ that rotates the spin frame, we can use average Hamiltonian theory (AHT), where the time-dependent Hamiltonian $H(t)$ is approximated by a time-independent Hamiltonian $H_{\mathrm{Floq}}$ (*23*). Each $i_{\mathrm{th}}$ pulse is rotating the spin-frame such that the system evolves under the action of a rotated-frame Hamiltonian $\tilde{\mathcal{H}}_i$ during the propagation time $\tau_i$ (Fig. 1B, top) (*14*); a thorough derivation can be found in (*23*). After an integer number of Floquet cycles, the resulting zeroth order Floquet Hamiltonian is $H_{\mathrm{Floq}} = \frac{1}{t_c} \sum_{i=1}^{5} \tilde{\mathcal{H}}_i \tau_i$ and, for our specific implementation, reads

$$H_{\mathrm{XYZ}} = \frac{2}{3} \sum_{i,j} J_{ij}/\hbar \left( \delta_x S_x^i S_x^j + \delta_y S_y^i S_y^j + \delta_z S_z^i S_z^j \right) \qquad (2)$$

where $\delta_x = 1 - v + u$, $\delta_y = 1 + w - u$, and $\delta_z = 1 - w + v$ are the control parameters allowing one to transform the natural XX Hamiltonian into an effective XYZ form. According to AHT, this description is valid as long as the period of the pulse sequence—that is, the cycling time $t_c$—is much shorter than the typical time scales of the mutual interactions $J_{ij}$, as expressed by the condition

$$J_{ij} \cdot t_c \ll 2\pi \qquad (3)$$

In our experimental implementation, the spin is represented by the two Rydberg states $|48S_{1/2}, m_j = 1/2\rangle = |\downarrow\rangle$ and $|48P_{3/2}, m_j = 1/2\rangle = |\uparrow\rangle$ in an ultracold gas of $^{87}$Rb atoms. The two states couple through a dipole-dipole interaction, yielding coefficients $J_{ij} = 2\, C_3(\theta_{ij})/r_{ij}^3$ in Eq. 1, where $C_3(\theta_{ij})$ is the angle-dependent dipolar coupling parameter and $r_{ij}$ represents the spatial separation between atoms $i$ and $j$ (*17*, *23*). Our system of a few hundred spins is disordered owing to the random positions of the $N$ Rydberg atoms in the cloud. The mean value of the interaction strength, defined as $J_{\mathrm{m}} = 1/N \sum_i \left| \sum_j J_{ij} \right|$, can be modified by changing the Rydberg density. A microwave field that resonantly couples the two Rydberg states allows us to globally manipulate the spins. The field is controlled by an arbitrary waveform generator (AWG) with adjustable amplitude and phase (*23*). The system's total magnetization $M = \sqrt{M_x^2 + M_y^2 + M_z^2}$ is mapped out by a tomographic measurement (*23*).

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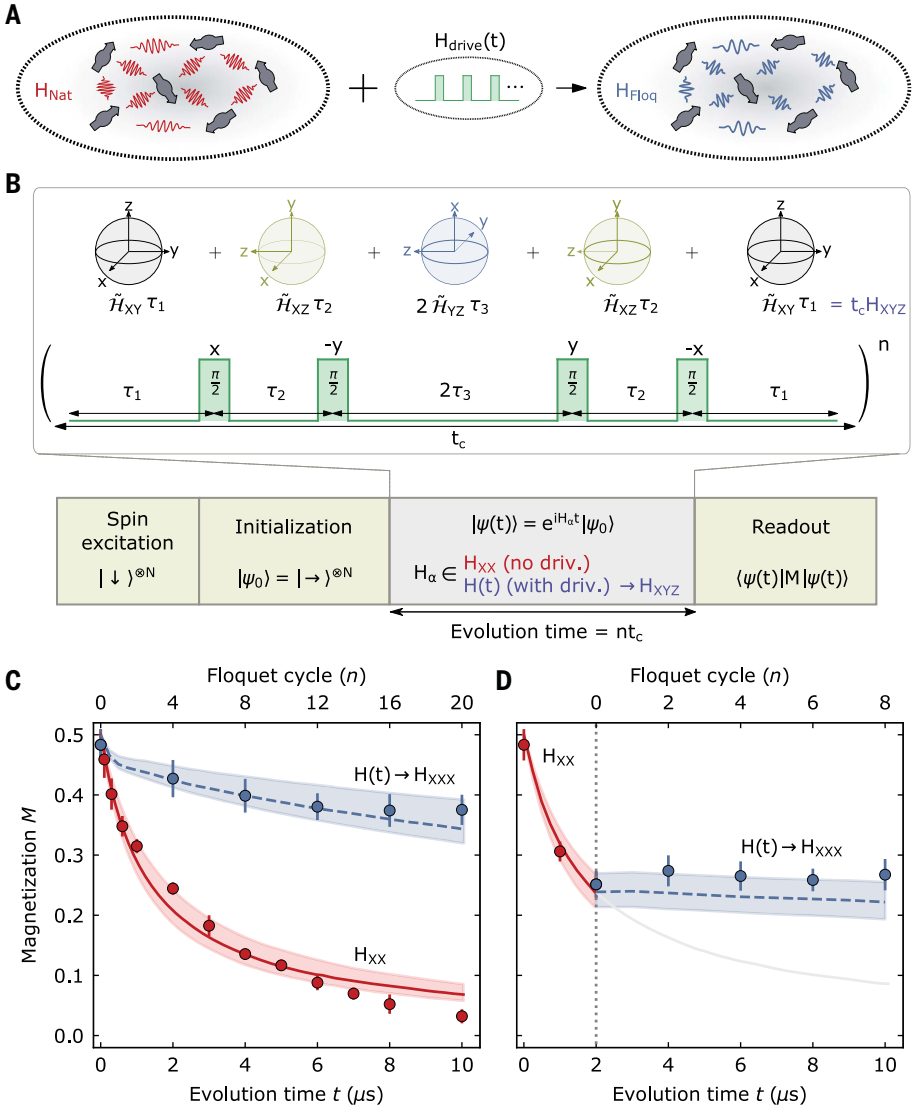



**Fig. 1. Hamiltonian engineering of an isolated spin system.** (**A**) A periodically applied Hamiltonian $H_{\mathrm{drive}}(t)$ is used to transform the natural Hamiltonian $H_{\mathrm{Nat}}$ of the system into a time-independent Floquet Hamiltonian $H_{\mathrm{Floq}}$. In this work, we have chosen $H_{\mathrm{Nat}}$ to be represented by a Heisenberg XX Hamiltonian $H_{\mathrm{XX}}$. The transformation of the Hamiltonian is illustrated by the different connection lines between the spins in the left and right panels. (**B**) Experimental protocol for measuring the total magnetization of the system under periodic driving composed by $n$ repetitions of four $\pi/2$ pulses in different directions ($x$, $-y$, $y$, $-x$) with adjustable delay time. The spheres indicate the spin-frame transformations, with the resulting piece-wise constant rotated-frame Hamiltonians $\tilde{\mathcal{H}}_{\alpha\beta} = \sum_{i,j} J_{ij}/\hbar\left(\tilde{S}^i_\alpha \tilde{S}^j_\alpha + \tilde{S}^i_\beta \tilde{S}^j_\beta\right)$ shown below. (**C**) Magnetization dynamics obtained by the natural Hamiltonian $H_{\mathrm{XX}}$ (red points) and by the driven Hamiltonian $H(t)$ for $t_c = 0.5$ μs (blue points) and $J_m/2\pi = 0.2$ MHz. (**D**) Magnetization dynamics when the Floquet cycles start after 2 μs for $t_c = 1$ μs and $J_m/2\pi = 0.2$ MHz. Lines are dTWA simulations of the respective dynamics under $H_{\mathrm{XX}}$ (solid) and $H(t)$ (dashed), including uncertainties of the Rydberg density (shaded areas) (*23*). Error bars are the standard error of the mean resulting from 100 repetitions of the experiment.

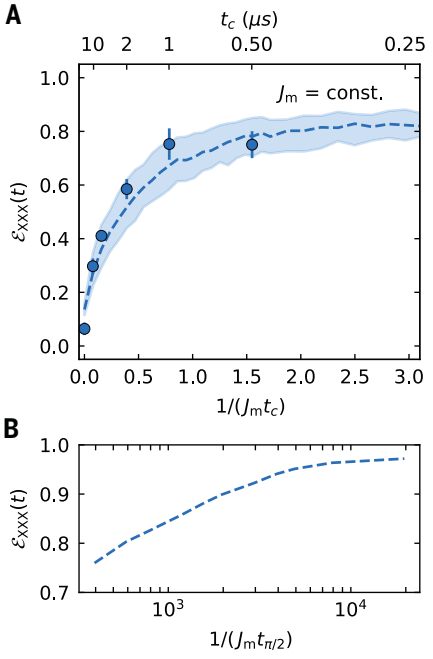



**Fig. 2. Efficiency of engineering the target Hamiltonian.** (**A**) Efficiency at $t = 10$ μs as a function of $1/(J_m t_c)$ for a fixed mean interaction strength $J_m/2\pi = 0.2$ MHz and $t_{\pi/2} = 12.5$ ns. Dashed lines are dTWA simulations of $H(t)$ with the respective sequence parameters and density uncertainties (shaded area). Error bars are the standard error of the mean resulting from 100 repetitions of the experiment. (**B**) Numerical dTWA simulations of the efficiency at $t = 10$ μs with pulse widths ranging from $t_{\pi/2} = 12.5$ to 0.25 ns for $t_c = 50$ ns and $J_m/2\pi = 0.2$ MHz. See (*23*) for details.

In a first series of experiments, we choose the pulse sequence such that $u = v = w$, which is formally equivalent to the so-called WAHUHA sequence (*12*) widely used in NMR to realize dynamical decoupling and to suppress spin-spin interactions. For the isolated quantum system considered here, this particular sequence implies $\delta_x = \delta_y = \delta_z$, resulting in the engineering of a symmetric Floquet Hamiltonian $H_{\mathrm{Floq}} = H_{\mathrm{XXX}}$, where the total spin constitutes a conserved quantity (*23*).

The experimental protocol, as shown in Fig. 1B, starts with the excitation of the $|\downarrow\rangle^{\otimes N}$ Rydberg state, followed by a $\pi/2$ pulse, thus initializing the spins in the product state $|\rightarrow\rangle_x^{\otimes N} = 1/\sqrt{2}(|\downarrow\rangle+|\uparrow\rangle)^{\otimes N}$ (total magnetization pointing along the $x$ direction). The spin system then evolves under the Hamiltonian given by Eq. 1. After time $t$, the total magnetization is measured.

Without periodic driving, the dynamics is governed by $H_{\mathrm{XX}}$, resulting in a fast relaxation toward a demagnetized state within ~10 μs (red points in Fig. 1C). In stark contrast, we observe a pronounced slowing down of the relaxation dynamics when we apply the WAHUHA sequence with $H_{\mathrm{XXX}}$ as the target Hamiltonian (blue points in Fig. 1C). To confirm that the stalling of the dynamics is independent of the initial state, we let the system first evolve under $H_{\mathrm{XX}}$ for 2 μs to create an entangled state. We then apply the WAHUHA sequence and find a complete freezing out of the magnetization dynamics (see Fig. 1D), as indeed expected for the symmetric $H_{\mathrm{XXX}}$.

To understand why the evolution in Fig. 1C is not entirely frozen, we have to consider the disordered nature of the system. The distance between a pair of Rydberg atoms has a lower bound $r_b$, determined by the so-called Rydberg

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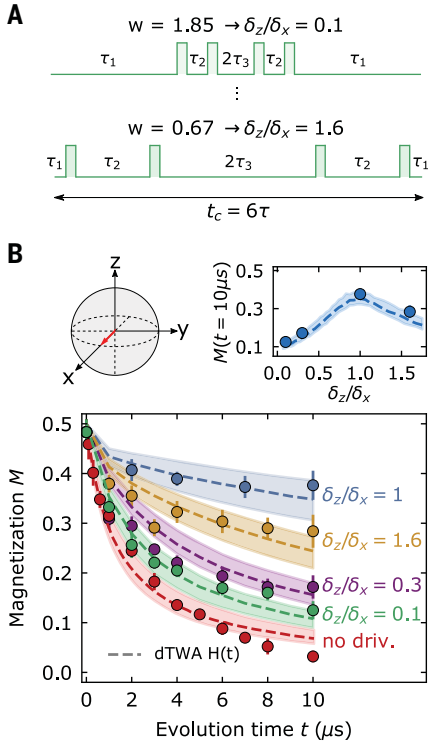



**Fig. 3. Engineered out-of-equilibrium dynamics.** (**A**) Illustration of the pulse sequence for a XXZ Hamiltonian for an anisotropy parameter $\delta_z/\delta_x$ = 0.1 and 1.6, respectively, with fixed $u = (w + v)/2$ and $v$ = 1. (**B**) Magnetization dynamics for different pulse delays, with $t_c = 1$ μs, $t_{\pi/2} = 12.5$ ns, and $J_m/2\pi = 0.2$ MHz (bottom). The shaded area corresponds to the density uncertainty in dTWA simulations of $H(t)$ (*23*), and error bars are the standard error of the mean resulting from 100 repetitions of the experiment. The red arrow on the Bloch sphere (top left) indicates the initial state. Magnetization at $t$ = 10 μs as function of the anisotropy parameter $\delta_z/\delta_x$ is shown at the top right.

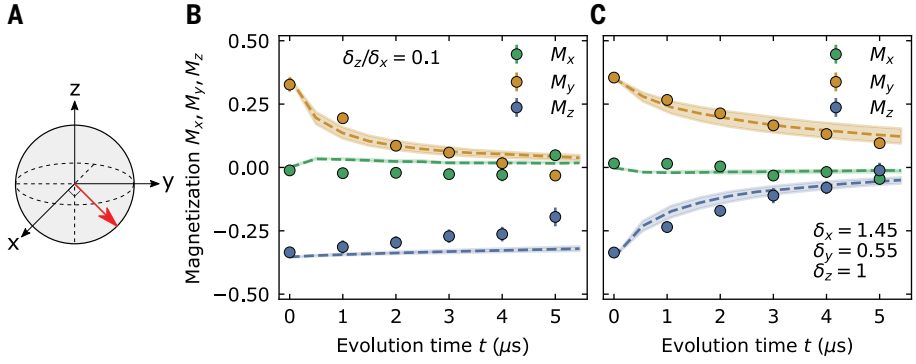



**Fig. 4. Consecutive symmetry breaking for a dual-component initial state.** (**A**) Representation of the state initialized with a $\pi/4$ pulse on the Bloch sphere. (**B**) Dynamics of the magnetization components for a XXZ Hamiltonian. (**C**) Dynamics of the magnetization component for a fully anisotropic XYZ Hamiltonian. In (B) and (C), parameters are $t_c = 0.5$ μs, $t_{\pi/2} = 10.7$ ns, and $J_m/2\pi = 0.4$ MHz, and the shaded areas correspond to the density uncertainty of dTWA simulations of $H(t)$ (dashed lines) (*23*).

blockade induced by the laser excitation process (*24*), resulting in a maximum interaction strength of $J_{max}/2\pi = 2\ C_3/(2\pi r_b^3) = 18$ MHz. For the closest pairs, this turns out to be large compared to the inverse cycling time $1/t_c = 2$ MHz, thus violating the condition required by Eq. 3 for the validity of an effective time-independent Hamiltonian description. The result is a slow remnant relaxation of the total magnetization. In Fig. 1D, however, the strongest interacting pairs of spins have demagnetized before the drive is applied and thus no longer contribute to the total magnetization. This qualitative interpretation is confirmed by a semiclassical simulation based on the discrete truncated Wigner approximation method (dTWA) (*25*) for the full time-dependent Hamiltonian given by Eq. 1, which quantitatively reproduces all essential features of the magnetization dynamics.

To quantify experimental deviations from ideal FHE, we introduce an efficiency for engineering the target Hamiltonian $H_{XXX}$ defined as $\mathcal{E}_{XXX}(t) = \frac{M(t)}{1/2}$, where $M(t)$ is the observed magnetization after a time $t$. An efficiency of $\mathcal{E}_{XXX}(t) \equiv 1$ corresponds to the dynamics expected for perfectly matching an XXX Hamiltonian. Figure 2A shows the efficiency as a function of $1/(J_m t_c)$ with varying $t_c$ and fixed $J_m$. If no pulse sequence is applied, which corresponds to $t_c$ reaching infinity, the efficiency is close to zero, owing to the relaxation of magnetization under the action of a pure $H_{XX}$, as shown in Fig. 1C. For shorter $t_c$, a growing number of spin pairs fulfill the condition of Eq. 3, resulting in an increase in the efficiency. Eventually, for the shortest achievable cycling times, the efficiency approaches a finite value in good quantitative agreement with the dTWA prediction.

That the efficiency does not reach unity under our experimental conditions can be attributed to the influence of interactions during the application of the pulses. This influence would become negligible if the pulse length $t_{\pi/2}$ was reduced beyond the current shortest value given by the bandwidth of our AWG. In Fig. 2B, we determine the reachable efficiency by numerically simulating pulse lengths down to $t_{\pi/2} = 0.25$ ns at fixed $1/(J_m t_c) = 16$. Under these conditions, which are within reach with today's cutting-edge microwave technology, the efficiency would indeed approach unity.

To demonstrate the full potential of FHE for programmable quantum simulation, we now proceed to a larger range of Hamiltonians that possess different symmetries and explore the resulting dynamics for different initial states. The freezing of the magnetization under the engineered XXX Hamiltonian observed in the previous set of measurements (see Fig. 1, C and D) is a direct consequence of the $SU(2)$ symmetry in the system. Using the pulse sequence shown in Fig. 3A, we can break this symmetry by setting $\delta_x = \delta_y \neq \delta_z$, effectively engineering a XXZ Hamiltonian that does not conserve the magnetization for a state initialized in the $x$ direction.

Figure 3B shows the time evolution of the magnetization for different values of the anisotropy parameter $\delta_z/\delta_x$, ranging from 0.1 to 1.6. Depending on the choice of $\delta_z/\delta_x$, the magnetization relaxes on different time scales in a nonexponential fashion typical for a disordered system consisting of hundreds of spins (*26*). A noticeable feature is a nonmonotonous change of the observed relaxation dynamics as a function of $\delta_z/\delta_x$, which is well reproduced by the dTWA simulation of the time-dependent Hamiltonian (see inset of Fig. 3B) and in full agreement with the expectation for the time-independent target XXZ Hamiltonian [see (*23*)].

The XXZ Hamiltonian still features a $U(1)$ symmetry, which results in a conservation of the magnetization along the $z$ direction. To illustrate that the conservation is indeed a property of the Hamiltonian and not of the initial state, we initialize the system by a $\pi/4$ pulse, resulting in nonzero magnetization components along the $y$ and $z$ directions (see Fig. 4A). As shown in Fig. 4B, the time evolution clearly differs for each of the magnetization components. In particular, the $y$ magnetization relaxes, whereas the $z$ magnetization shows no relevant relaxation as a consequence of the $U(1)$ symmetry. Breaking this remaining $U(1)$ symmetry, by setting the pulse sequence to $\delta_x \neq \delta_y \neq \delta_z$ (fully anisotropic XYZ Hamiltonian), induces a nonconservation of the magnetization in the $z$ direction, resulting in a relaxation of all magnetization components (Fig. 4C). The $z$ magnetization relaxes faster than the $y$ magnetization because the dynamics of the former

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scales with $\delta_x - \delta_y = 0.9$, whereas the dynamics of the latter scales with $\delta_x - \delta_z = 0.45$ [for more details, see (*23*)].

FHE of isolated spin systems enables quantum simulation of many-body systems beyond equilibrium. The ability to synthesize a variety of many-body Hamiltonians allows the experimental exploration of important concepts such as spin transport (*27*), the generalization of fluctuation dissipation relations (*28*) to out-of-equilibrium systems, and the nature of thermalization mechanisms (*29*). For disordered systems, the FHE technique can be combined with the realization of time-reversal operations (*13*), with the aim of measuring out-of-time-order correlators; this will facilitate experimental investigation of the role of entanglement and information scrambling in out-of-equilibrium systems (*30*, *31*) as well as of nonergodic many-body localized states (*32*) in various spin Hamiltonians. The approach can be directly combined with current developments, for example, in preparing Rydberg atom arrays (*17*); this synthesis of efficient engineering of the Hamiltonian with precise control over the spatial arrangement provides a viable scenario for the realization of fully programmable quantum spin simulators.

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## SUPPLEMENTARY MATERIALS

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Methods
Supplementary Text
Figs. S1 to S3
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# Floquet Hamiltonian engineering of an isolated many-body spin system

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**Shaping the Hamiltonian**

Ultracold atoms have proven to be a versatile setting for simulating more complicated quantum systems. However, the type of the simulated Hamiltonian usually cannot be changed once the system has been engineered, although the parameters can be tuned. Geier *et al*. used rubidium-87 atoms in two excited Rydberg states to create a system in which the type of the Hamiltonian could be modified by periodic driving. The researchers started with a system that was initially described by the so-called Heisenberg XX Hamiltonian. By using a periodic series of pulses, the Hamiltonian was transformed into a different target Hamiltonian, which was corroborated by monitoring the system's dynamics. —JS

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